

Independent System Operator (ISO) / Market Operators		Gather Input & Prepare Collect system, bids, and grid state data		Where AI Helps
Independent System Operator (ISO) / Market Operators (Focused on system-wide efficiency, congestion, reliability, and market design) System Operator / Dispatcher (RT) Real-Time Market Operator Day-Ahead Market Operator Market Analyst Transmission Planner Reliability Engineer Market Design / Policy Analyst	Real Time Real-time dispatch & system balance report Congestion & transmission constraint report Frequency & reserve monitoring report Imbalance / shortage risk report Real-time LMP / price signal report	Goal	Maintain real-time system balance (supply = demand) Ensure grid reliability and manage congestion Detect and respond to deviations quickly	Real-time data validation
		Actions	Collect real-time data (load, generation, outages, transmission flows) Monitor system conditions (frequency, congestion, reserves) Validate actual vs forecast and identify deviations	
		Thoughts	“Is the system balanced right now?” “Are there any congestion or reliability risks?” “Do we need immediate intervention?”	Example AI Capability AI detects abnormal grid data / missing bids
		Pain Point	Incomplete or delayed real-time system data Sudden outages or renewable variability Managing multiple constraints simultaneously	
		Emotions	High alert during system fluctuations Pressure to maintain stability continuously Relief when grid is balanced and stable	
		Tool	SCADA / EMS systems (grid monitoring) Real-time market systems (RTM platforms) PLEXOS / simulation tools for validation	
	Day Ahead / Forward Market clearing report (LMP, dispatch, unit commitment) Congestion & nodal pricing analysis Resource adequacy / reserve margin report Transmission constraint & flow report Market efficiency / price behaviour report	Goal	Prepare accurate inputs for day-ahead market clearing Ensure system reliability under forecasted conditions Support efficient and fair market operation	
		Actions	Gather forecasts (load, renewables, outages, transmission constraints) Validate bids/offers from generators and demand participants Prepare inputs for market clearing simulation (DAM)	
		Thoughts	“Are forecasts realistic and complete?” “Do bids/offers reflect actual system capability?” “Will the system remain reliable under this plan?”	
		Pain Point	Forecast uncertainty (load, wind, solar) Inconsistent or strategic bidding behaviour Complexity of modelling congestion and constraints	
		Emotions	Analytical during preparation phase Concern about forecast accuracy Confidence when inputs are validated	
		Tool	Market management systems (DAM platforms) Forecasting tools (load, renewable, outages) PLEXOS / market simulation tools	

Independent System Operator (ISO) / Market Operators		Simulation Setting Configure market clearing & constraints		Where AI Helps
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		Actions	Set short-term horizon (intra-hour / hourly) and resolution Apply real-time inputs (load, generation, outages, transmission limits, reserves) Select outputs (LMP, dispatch, congestion, reserve activation, imbalance)	
		Thoughts	“Are all real-time constraints correctly captured?” “Is the model granular enough for current operations?” “Will outputs reflect actual system behaviour?”	
		Pain Point	Incorrect horizon or resolution reduces usefulness of results Missing constraints can lead to unrealistic dispatch outcomes Too many outputs make it hard to focus on critical signals	
		Emotions	Careful while configuring high-impact real-time runs Pressure due to time-sensitive decisions Confidence when setup reflects actual grid conditions	
		Tool	PLEXOS / real-time dispatch simulation SCADA / EMS inputs (grid state, outages, flows) RTM (Real-Time Market) systems	
		Goal	Configure market clearing simulation for DAM and forward planning Incorporate all bids/offers and system constraints accurately Generate outputs for efficient, reliable market outcomes	
		Actions	Define horizon (day-ahead, week-ahead) and time resolution Input bids/offers, demand forecasts, outages, and transmission constraints Select outputs (LMP, unit commitment, congestion, reserve margins)	
	Day Ahead / Forward Market clearing report (LMP, dispatch, unit commitment) Congestion & nodal pricing analysis Resource adequacy / reserve margin report Transmission constraint & flow report Market efficiency / price behaviour report	Thoughts	“Is the market model capturing all constraints correctly?” “Are bids/offers aligned with system capability?” “Will this produce a reliable and efficient market outcome?”	
		Pain Point	Complexity of integrating bids, constraints, and forecasts Risk of incorrect assumptions impacting market outcomes Difficulty ensuring model reflects real-world system behaviour	
		Emotions	Analytical during setup of market model Concern about correctness of assumptions Confidence when model setup is validated	
		Tool	PLEXOS market simulation (DAM / forward runs) Market Management Systems (bids/offers handling) Forecasting & constraint modelling tools	

Independent System Operator (ISO) / Market Operators		Execute Back cast Model & RUN Validate market behaviour vs history		Where AI Helps
Independent System Operator (ISO) / Market Operators (Focused on system-wide efficiency, congestion, reliability, and market design) System Operator / Dispatcher (RT) Real-Time Market Operator Day-Ahead Market Operator Market Analyst Transmission Planner Reliability Engineer Market Design / Policy Analyst	Real Time Real-time dispatch & system balance report Congestion & transmission constraint report Frequency & reserve monitoring report Imbalance / shortage risk report Real-time LMP / price signal report	Goal	Reproduce historical system conditions to validate real-time dispatch behaviour Verify congestion, imbalance, and reserve activation patterns Improve confidence in real-time operational models	Market pattern learning
		Actions	Load historical data (load, generation, outages, transmission flows, prices) Run back cast for selected time periods (critical events, peak hours) Compare simulated vs actual outcomes (dispatch, congestion, imbalance)	
		Thoughts	“Does the model replicate actual system behaviour?” “Where are the deviations between model and reality?” “Can this model be trusted for real-time operations?”	
		Pain Point	Historical data gaps or inconsistencies Difficult to capture real-time events (outages, congestion spikes) Hard to isolate root cause of mismatch (data vs model vs constraints)	
		Emotions	Focused during validation of critical periods Frustrated when model deviates from actual behaviour Confident when outputs match historical outcomes	
		Tool	PLEXOS back casting / real-time simulation Historical SCADA / EMS / RTM data Market and congestion history datasets	
	Day Ahead / Forward Market clearing report (LMP, dispatch, unit commitment) Congestion & nodal pricing analysis Resource adequacy / reserve margin report Transmission constraint & flow report Market efficiency / price behaviour report	Goal	Validate market clearing logic against historical DAM outcomes Ensure model correctly captures price formation and congestion Build a reliable benchmark for future simulations and market design	Example AI Capability AI learns price/congestion patterns and detects deviations
		Actions	Run back cast using historical bids/offers, demand, and constraints Compare simulated vs actual DAM results (LMP, unit commitment, congestion) Adjust model assumptions and rerun until alignment is achieved	
		Thoughts	“Is the market model producing realistic prices and dispatch?” “Are congestion and constraints captured correctly?” “Is this accurate enough for future market simulations?”	
		Pain Point	Complexity of modelling bids/offers and market behaviour Difficulty capturing strategic participant behaviour Need high accuracy for regulatory and market validation	
		Emotions	Analytical while benchmarking model vs actual Concern when results diverge significantly Confidence when model aligns with historical market outcomes	
		Tool	PLEXOS market back casting / DAM simulation Historical market data (bids/offers, prices, demand) Data pipelines and validation tools	

Independent System Operator (ISO) / Market Operators		Analyse Solution Analyse reliability, congestion, and prices		Where AI Helps
Independent System Operator (ISO) / Market Operators (Focused on system-wide efficiency, congestion, reliability, and market design) System Operator / Dispatcher (RT) Real-Time Market Operator Day-Ahead Market Operator Market Analyst Transmission Planner Reliability Engineer Market Design / Policy Analyst	Real Time Real-time dispatch & system balance report Congestion & transmission constraint report Frequency & reserve monitoring report Imbalance / shortage risk report Real-time LMP / price signal report	Goal	Evaluate system performance under current operating conditions Identify congestion, imbalance, and reliability risks Ensure dispatch and reserves are maintaining system stability	Root cause analysis Example AI Capability “Congestion driven by Line X overload + wind drop”
		Actions	Analyse outputs (frequency, dispatch, reserves, congestion, imbalance) Compare expected vs actual system behaviour Flag issues and trigger corrective operational actions	
		Thoughts	“Is the system operating within safe limits?” “Are there any emerging congestion or imbalance risks?” “Do we need immediate intervention?”	
		Pain Point	High volume of outputs makes analysis difficult Real-time issues evolve quickly and are hard to track Linking results directly to root causes is challenging	
		Emotions	Alert while monitoring system performance Pressure to identify and act on risks quickly Relief when system remains stable	
		Tool	SCADA / EMS visualization systems Real-time market monitoring dashboards PLEXOS output analysis tools	
	Day Ahead / Forward Market clearing report (LMP, dispatch, unit commitment) Congestion & nodal pricing analysis Resource adequacy / reserve margin report Transmission constraint & flow report Market efficiency / price behaviour report	Goal	Evaluate market clearing results (prices, dispatch, congestion) Assess system reliability and adequacy under forecasted conditions Identify inefficiencies or risks in market outcomes	
		Actions	Analyse outputs (LMP, unit commitment, congestion, reserve margins) Evaluate impact of constraints on price and dispatch Identify trends, anomalies, and system bottlenecks	
		Thoughts	“Are prices and dispatch reflecting true system conditions?” “Where are the major congestion points?” “Is the system reliable under this plan?”	
		Pain Point	Complex relationship between constraints, prices, and dispatch Difficulty interpreting congestion and price signals Need to validate results for regulatory and market fairness	
		Emotions	Analytical while reviewing system and market outcomes Uncertainty when results are complex or unclear Confidence when patterns and drivers are understood	
		Tool	PLEXOS output reports & congestion analysis tools Market analytics dashboards BI / visualization tools for trend analysis	

Independent System Operator (ISO) / Market Operators		Forecast/Base Model & Run Forecast system conditions & market outcomes		Where AI Helps
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		Actions	Run short-term forecasts using updated load, renewable, and outage inputs Update base model with current system constraints and network conditions Generate forward outputs (dispatch, congestion risk, reserve requirements)	
		Thoughts	“What will system conditions look like in the next few hours?” “Are there any upcoming congestion or reliability risks?” “Do we need preventive actions before imbalance occurs?”	
		Pain Point	Forecast uncertainty due to renewable variability Rapidly changing system conditions reduce prediction accuracy Limited visibility into unexpected outages	
		Emotions	Alert while anticipating near-term risks Uncertain due to forecast variability Confident when predictions align with trends	
		Tool	PLEXOS short-term forecasting / dispatch models SCADA / EMS real-time inputs Weather and renewable forecast systems	
	Day Ahead / Forward Market clearing report (LMP, dispatch, unit commitment) Congestion & nodal pricing analysis Resource adequacy / reserve margin report Transmission constraint & flow report Market efficiency / price behaviour report	Goal	Forecast system conditions for day-ahead market clearing Ensure reliability and efficient price formation Support market design and policy evaluation	Example AI Capability AI predicts congestion hotspots and reserve shortages
		Actions	Run base model using forecasted demand, bids/offers, outages, and constraints Generate outputs (LMP, unit commitment, congestion, reserve margins) Produce forecasts for multiple conditions (demand, renewables, fuel mix)	
		Thoughts	“Will this forecast lead to efficient and reliable market outcomes?” “Are price signals reflecting system constraints correctly?” “Is the system secure under these predicted conditions?”	
		Pain Point	High uncertainty in demand and renewable forecasts Complexity in modelling bids, constraints, and congestion together Ensuring forecasts are accurate enough for market clearing	
		Emotions	Analytical while reviewing forecast outputs Concern about uncertainty in assumptions Confidence when model outputs are consistent	
		Tool	PLEXOS market forecasting & simulation Market Management Systems (DAM inputs) Forecasting tools (load, renewable, outages, constraints)	

Independent System Operator (ISO) / Market Operators		Forecast/Base Model + Scenario-A,,, & Run Test system stress & market scenarios		Where AI Helps
Independent System Operator (ISO) / Market Operators (Focused on system-wide efficiency, congestion, reliability, and market design) System Operator / Dispatcher (RT) Real-Time Market Operator Day-Ahead Market Operator Market Analyst Transmission Planner Reliability Engineer Market Design / Policy Analyst	Real Time Real-time dispatch & system balance report Congestion & transmission constraint report Frequency & reserve monitoring report Imbalance / shortage risk report Real-time LMP / price signal report	Goal	Evaluate system behaviour under multiple short-term risk scenarios Identify potential reliability issues before they occur Prepare contingency actions for real-time operations	Stress scenario generation Example AI Capability AI simulates extreme events (outages, demand spikes)
		Actions	Create short-term scenarios (load spike, outage event, low renewable output) Run simulations using updated system conditions and constraints Compare outputs (congestion, reserves, imbalance risk) across scenarios	
		Thoughts	“What happens if system conditions suddenly worsen?” “Which scenario poses the highest reliability risk?” “Do we have a contingency plan ready?”	
		Pain Point	Limited time to create and evaluate multiple scenarios Difficulty identifying most critical risk scenarios High variability in results across scenarios	
		Emotions	Alert while preparing for uncertainty Pressure to evaluate risks quickly Confidence when contingency plans are clear	
		Tool	PLEXOS short-term scenario simulation SCADA / EMS inputs (real-time system state) Real-time monitoring and forecasting tools	
	Day Ahead / Forward Market clearing report (LMP, dispatch, unit commitment) Congestion & nodal pricing analysis Resource adequacy / reserve margin report Transmission constraint & flow report Market efficiency / price behaviour report	Goal	Test market outcomes under multiple future scenarios Evaluate system reliability and congestion under uncertainty Support market design, policy, and planning decisions	
		Actions	Define scenarios (demand variation, renewable variability, outages, policy changes) Run simulations with bids/offers, constraints, and network conditions Generate outputs (LMP, congestion, unit commitment, reserve margins)	
		Thoughts	“How will the system behave under different future conditions?” “Are there scenarios where reliability is at risk?” “Do market outcomes remain efficient across scenarios?”	
		Pain Point	Managing large number of complex scenarios Difficulty interpreting impact of constraints across scenarios Ensuring scenario realism and completeness	
		Emotions	Analytical while evaluating system behaviour Uncertainty when scenarios diverge significantly Confidence when consistent patterns emerge	
		Tool	PLEXOS scenario modelling & simulation Market Management Systems (DAM + planning inputs) Forecasting, policy, and constraint modelling tools	

Independent System Operator (ISO) / Market Operators		Compare Scenarios Select most stable & efficient system outcome		Where AI Helps
Independent System Operator (ISO) / Market Operators (Focused on system-wide efficiency, congestion, reliability, and market design) System Operator / Dispatcher (RT) Real-Time Market Operator Day-Ahead Market Operator Market Analyst Transmission Planner Reliability Engineer Market Design / Policy Analyst	Real Time Real-time dispatch & system balance report Congestion & transmission constraint report Frequency & reserve monitoring report Imbalance / shortage risk report Real-time LMP / price signal report	Goal	Identify scenario with lowest reliability risk for immediate operation Compare system stress across different short-term conditions Select best contingency plan to maintain system balance	Risk ranking
		Actions	Compare scenario outputs (congestion, reserves, imbalance, frequency risk) Identify worst-case vs most stable system conditions Select and prepare corrective or preventive operational actions	
		Thoughts	“Which scenario puts the system at highest risk?” “Are we prepared for worst-case conditions?” “Which scenario should guide immediate action?”	
		Pain Point	Limited time to evaluate multiple scenarios Small differences can significantly impact reliability Difficulty prioritizing critical risk scenarios	
		Emotions	High alert while assessing system risk Pressure to make fast, accurate decisions Confidence when a clear safe scenario is identified	
		Tool	PLEXOS scenario comparison outputs SCADA / EMS real-time visualization Reliability and congestion monitoring dashboards	
		Goal	Identify scenario that ensures reliable and efficient market outcomes Compare trade-offs between congestion, cost, and reliability Support market design, policy, and planning decisions	
		Actions	Compare outputs (LMP, congestion patterns, unit commitment, reserves) Analyse sensitivity across demand, renewable, and outage scenarios Select scenario for market clearing, planning, or policy evaluation	
	Day Ahead / Forward Market clearing report (LMP, dispatch, unit commitment) Congestion & nodal pricing analysis Resource adequacy / reserve margin report Transmission constraint & flow report Market efficiency / price behaviour report	Thoughts	“Which scenario ensures both reliability and efficient pricing?” “Where does congestion create inefficiencies?” “Are results consistent across key assumptions?”	Example AI Capability AI ranks scenarios: “Scenario B has highest reliability risk”
		Pain Point	Complexity in interpreting price vs constraint relationships Managing large number of scenarios Ensuring fairness and realism in scenario comparison	
		Emotions	Analytical during comparison Uncertainty when results conflict Confidence when patterns and optimal scenario are clear	
		Tool	PLEXOS scenario comparison & reporting Market analytics dashboards BI / visualization tools for multi-scenario analysis	

Independent System Operator (ISO) / Market Operators		Results Clear market and ensure system reliability		Where AI Helps
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		Actions	Monitor actual system performance (frequency, flows, reserves, congestion) Execute corrective actions (redispatch, reserve activation, load balancing) Continuously adjust operations based on real-time conditions	
		Thoughts	“Is the system stable after interventions?” “Are all constraints being respected?” “Do we need further adjustments?”	
		Pain Point	Continuous need for monitoring and rapid response Unexpected deviations from predicted conditions Difficulty linking outcomes to specific actions or causes	
		Emotions	Relief when system remains stable Stress during unexpected disturbances Confidence when operations are under control	
		Tool	SCADA / EMS operational systems Real-time monitoring dashboards Control room tools for dispatch and reserve activation	
	Day Ahead / Forward Market clearing report (LMP, dispatch, unit commitment) Congestion & nodal pricing analysis Resource adequacy / reserve margin report Transmission constraint & flow report Market efficiency / price behaviour report	Goal	Deliver efficient and reliable market outcomes (price, dispatch, congestion) Ensure fairness and transparency in market clearing Provide insights for future planning and market design improvements	Example AI Capability AI alerts: “Frequency risk in next 15 min”
		Actions	Publish market results (LMP, unit commitment, congestion outcomes) Communicate outcomes to participants and stakeholders Feed results into planning, policy, and future model improvements	
		Thoughts	“Are market outcomes efficient and fair?” “Did constraints drive expected price signals?” “What improvements are needed for future runs?”	
		Pain Point	Differences between forecast and actual outcomes Stakeholder scrutiny and regulatory requirements Complexity in explaining price and congestion outcomes	
		Emotions	Satisfaction when market clears efficiently Pressure from regulatory and stakeholder expectations Confidence when outcomes are transparent and defensible	
		Tool	PLEXOS output and reporting tools Market Management Systems (result publication) BI / reporting dashboards for stakeholders	